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Project
537-561 Johnson Avenue Project

Section
Prop. Central Column Pilecap (310k Service)

Calc. by
HSK

Date
3/22/2017

Chk'd by

Date

Job Ref.
170102

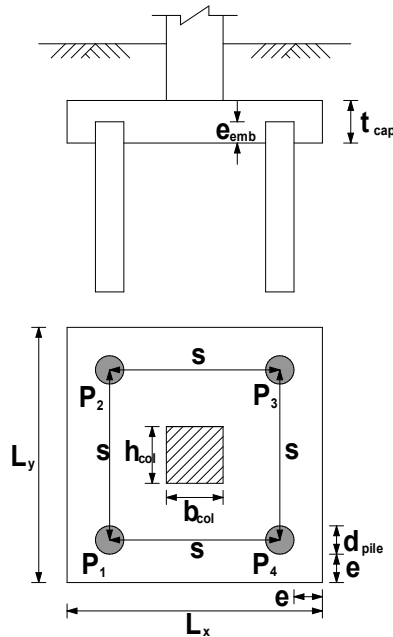
Sheet no./rev.
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App'd by
Hazim K.

Date
3/22/2017

PILE CAP DESIGN (ACI318-11)

TEDDS calculation version 1.1.06



Pile details

Number of piles

$$N = 4$$

Pile diameter

$$d_{\text{pile}} = 10.0 \text{ in}$$

Pile capacity (service)

$$P_{\text{cap}} = 65.0 \text{ kips}$$

Pile spacing provided

$$s = 36.0 \text{ in}$$

Minimum pile spacing required

$$s_{\text{min}} = \max(36 \text{ in}, (d_{\text{pile}} + 24 \text{ in})) = 36.0 \text{ in}$$

PASS - Provided spacing is not less than minimum required

Edge distance of pile from pile cap edge

$$e = 15.0 \text{ in}$$

Minimum edge distance required

$$e_{\text{min}} = 10 \text{ in}$$

PASS - Provided edge distance is not less than minimum required

Pile cap details

Concrete cover

$$c_c = 4.5 \text{ in}$$

Length of pile cap along X axis

$$L_x = s + d_{\text{pile}} + 2 \times e = 6.3 \text{ ft}$$

Length of pile cap along Y axis

$$L_y = 6.3 \text{ ft}$$

Overall pile cap thickness

$$t_{\text{cap}} = 44.0 \text{ in}$$

Pile embedment distance

$$e_{\text{emb}} = 8.0 \text{ in}$$

Minimum pile embedment distance

$$e_{c_{\text{min}}} = 6 \text{ in}$$

PASS - Provided embedment distance is not less than minimum required

Column details

Column width along X axis

$$b_{\text{col}} = 16.0 \text{ in}$$

Column width along Y axis

$$h_{\text{col}} = 16.0 \text{ in}$$

Material properties

Yield strength of tension reinforcement

$$f_y = 60 \text{ ksi}$$



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Compressive strength of concrete

$$f_c = 5 \text{ ksi}$$

Concrete density

$$\rho_c = 150 \text{ lb/ft}^3$$

Reinforcement details (ACI 318, cl. 7.12, cl. 10.5)

Diameter number of steel bar along X axis

$$D_{x_bar_num} = 9$$

Diameter of steel bar along X axis

$$D_{x_bar} = 1.13 \text{ in}$$

Effective depth of cap

$$d = t_{cap} - e_{emb} - C_c - D_{x_bar} / 2 = 30.9 \text{ in}$$

Total number of steel bars along X axis

$$N_{x_bar} = 11$$

Spacing of steel bars along X axis

$$S_{x_bar} = (L_y - 2 \times C_c - D_{x_bar}) / (N_{x_bar} - 1) = 6.6 \text{ in}$$

Area of steel provided along X axis

$$A_{st_x} = N_{x_bar} \times 0.25 \times \pi \times D_{x_bar}^2 = 11.0 \text{ in}^2$$

Gross cross sectional area of cap along Y axis

$$A_{g_y} = L_y \times t_{cap} = 3344.0 \text{ in}^2$$

C/S area of cap above bottom reinf. along Y axis

$$A_{b_y} = L_y \times d = 2351.1 \text{ in}^2$$

Minimum area of steel required along X axis

$$A_{st_min_x} = \max(0.0018 \times A_{g_y}, 200 \times A_{b_y} \times 1 \text{ psi} / f_y, 3 \times \sqrt{f_c \times 1 \text{ psi}} \times A_{b_y} / f_y)$$

$$A_{st_min_x} = 7.9 \text{ in}^2$$

PASS - Provided area of steel along X axis is greater than minimum required

Diameter number of steel bar along Y axis

$$D_{y_bar_num} = 9$$

Diameter of steel bar along Y axis

$$D_{y_bar} = 1.13 \text{ in}$$

Total number of steel bars along Y axis

$$N_{y_bar} = 11$$

Spacing of steel bars along Y axis

$$S_{y_bar} = (L_x - 2 \times C_c - D_{y_bar}) / (N_{y_bar} - 1) = 6.6 \text{ in}$$

Area of steel provided along Y axis

$$A_{st_y} = N_{y_bar} \times 0.25 \times \pi \times D_{y_bar}^2 = 11.0 \text{ in}^2$$

Gross cross sectional area of cap along X axis

$$A_{g_x} = L_x \times t_{cap} = 3344.0 \text{ in}^2$$

C/S area of cap above bottom reinf. along X axis

$$A_{b_x} = L_x \times d = 2351.1 \text{ in}^2$$

Minimum area of steel required along Y axis

$$A_{st_min_y} = \max(0.0018 \times A_{g_x}, 200 \times A_{b_x} \times 1 \text{ psi} / f_y, 3 \times \sqrt{f_c \times 1 \text{ psi}} \times A_{b_x} / f_y)$$

$$A_{st_min_y} = 7.9 \text{ in}^2$$

PASS - Provided area of steel along Y axis is greater than minimum required

Loading details

Strength reduction factor (flexure)

$$\phi_f = 0.90$$

Strength reduction factor (shear)

$$\phi_v = 0.75$$

Strength reduction factor (tied-arch)

$$\phi_t = 0.75$$

Column axial dead load

$$P_{dl} = 84.5 \text{ kips}$$

Area of pile cap

$$A_{cap} = L_x \times L_y = 40.1 \text{ ft}^2$$

Weight of pile cap

$$P_{wt} = (L_x \times L_y \times t_{cap} - 4 \times \pi \times d_{pile}^2 \times e_{emb} / 4) \times \rho_c = 21.8 \text{ kips}$$

Surcharge pressure on pile cap

$$\rho_{sur} = 0.1 \text{ ksf}$$

Surcharge load

$$P_{sur} = \rho_{sur} \times (L_x \times L_y - h_{col} \times b_{col}) = 5.7 \text{ kips}$$

Column axial live load

$$P_{ll} = 130.8 \text{ kips}$$

Total column load (unfactored)

$$P_n = P_{dl} + P_{ll} = 215.3 \text{ kips}$$

The service load per pile is

$$P_{n_pile} = (P_n + P_{wt} + P_{sur}) / 4 = 60.7 \text{ kips}$$

PASS - Pile capacity is sufficient

Total column load (factored)

$$P_u = 1.2 \times P_{dl} + 1.6 \times P_{ll} = 310.6 \text{ kips}$$

The ultimate load per pile is

$$P_{u_pile} = (P_u + 1.2 \times (P_{wt} + P_{sur})) / 4 = 85.9 \text{ kips}$$

Pile cap forces – tied arch method

$$w_c = \sqrt{((s / 2 - b_{col} / 4)^2 + (s / 2 - h_{col} / 4)^2)} = 19.8 \text{ in}$$

$$\theta = \text{atan}(d / w_c) = 57.4 \text{ deg}$$

$$\phi_1 = \min(\text{atan}((s / 2 - h_{col} / 4) / (s / 2 - b_{col} / 4)), \text{atan}((s / 2 - b_{col} / 4) / (s / 2 - h_{col} / 4)))$$



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	$\phi_1 = 45.0$ deg
Compression within pile cap	$C_{max} = P_{u_pile} / \sin(\theta) = 102.0$ kips
Tension within pile cap	$T = C_{max} \times \cos(\theta) \times \cos(\phi_1) = 38.9$ kips
Compression checks for pile cap	
Strength reduction factor (struct plain concrete)	$\phi_u = 0.55$
Strength reduction factor (bearing on concrete)	$\phi_b = 0.65$
Check diagonal comp. as an unreinforced column, using a core equivalent to 2 pile diameters (ACI 318-05, cl. 22.5.2)	
Bearing capacity of core equivalent to 2 pile dia	$P_c = \phi_u \times 0.60 \times f'_c \times \pi \times ((2 \times d_{pile})^2) / 4 = 466.5$ kips
PASS - Compression within cap is less than diagonal compression	
Check bearing capacity of pile cap over piles (ACI 318, cl. 22.5.5)	
Area of pile cap	$A_{cap} = 5776.00$ in ²
Cross sectional area of pile	$A_{pile} = \pi \times d_{pile}^2 / 4 = 78.54$ in ²
Bearing capacity of cap above pile	$P_{brg_pile} = \phi_u \times 0.85 \times f'_c \times A_{pile} \times \min(2, \sqrt{(A_{cap} / A_{pile})}) = 330.5$ kips
PASS - Bearing capacity pile cap over pile is sufficient	
Check bearing capacity of pile cap under column (ACI 318, cl. 10.14)	
Cross sectional area of column	$A_{col} = b_{col} \times h_{col} = 256.00$ in ²
Bearing capacity of cap below column	$P_{brg_col} = \phi_b \times 0.85 \times f'_c \times A_{col} \times \min(2, \sqrt{(A_{cap} / A_{col})}) = 1273.0$ kips
PASS - Bearing capacity pile cap under column is sufficient	
Area of steel check for moment resistance	
Distance from pile to face of column	$x_{face} = \max(s / 2 - b_{col} / 2, 0 \text{ in}) = 10.0$ in
Pressure due to weight and surcharge	$q_u = 1.2 \times (p_{sur} + P_{wt} / A_{cap}) = 0.8$ ksf
Ultimate moment at column face (Y axis)	$M_{uy} = 2 \times P_{u_pile} \times x_{face} - q_u \times L_y \times (L_x - b_{col})^2 / 8 = 126.7$ kip_ft
Net moment at column face (Y axis)	$M_{ny} = M_{uy} / \phi_f = 140.8$ kip_ft
	$R_{req_y} = M_{ny} / (f'_c \times d^2) / L_x = 0.005162$
Lever arm factor	$J_y = \sqrt{(0.25 - R_{req_y} / (0.85 \times 2))} + 0.5 = 0.9970$
Area of steel required to resist moment	$A_{st_req_Mx} = M_{ny} / (f_y \times J_y \times d) = 0.9$ in ²
Area of tension steel required for tied-arch behavior	$A_{st_req_T} = T / (\phi_t \times f_y) = 0.9$ in ²
Area of steel required along X axis	$A_{st_req_x} = \max(A_{st_req_Mx}, A_{st_req_T}) = 0.9$ in ²
PASS - Provided area of steel along X axis is sufficient	
Stress block depth factor	$\beta_1 = 0.83$
Depth of equivalent rectangular stress block	$a_x = A_{st_x} \times f_y / (0.85 \times f'_c \times L_x) = 2.27$ in
Balanced neutral axis depth(X axis)	$C_x = a_x / \beta_1 = 2.750$ in
Strain in steel (X axis)	$\epsilon_{ty_x} = 0.003 \times ((d - C_x) / C_x) = 0.03075$
Pass - Ductility is sufficient at ultimate strength	
Distance from pile to face of column	$y_{face} = \max(s / 2 - h_{col} / 2, 0 \text{ in}) = 10.0$ in
Ultimate moment at column face (X axis)	$M_{ux} = 2 \times P_{u_pile} \times y_{face} - q_u \times L_x \times (L_y - h_{col})^2 / 8 = 126.7$ kip_ft
Net moment at column face (X axis)	$M_{nx} = M_{ux} / \phi_f = 140.8$ kip_ft
	$R_{req_x} = M_{nx} / (f'_c \times d^2) / L_y = 0.005162$
Lever arm factor	$J_x = \sqrt{(0.25 - R_{req_x} / (0.85 \times 2))} + 0.5 = 0.9970$
Area of steel required to resist mmt along Y axis	$A_{st_req_My} = M_{nx} / (f_y \times J_x \times d) = 0.9$ in ²
Area of steel required along Y axis	$A_{st_req_y} = \max(A_{st_req_My}, A_{st_req_T}) = 0.9$ in ²
PASS - Provided area of steel along Y axis is sufficient	
Stress block depth factor	$\beta_1 = 0.8$



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Depth of equivalent rectangular stress block

$$a_y = A_{st_y} \times f_y / (0.85 \times f'_c \times L_y) = \mathbf{2.3 \text{ in}}$$

Balanced neutral axis depth(Y axis)

$$c_y = a_y / \beta_1 = \mathbf{2.8 \text{ in}}$$

Strain in steel (Y axis)

$$\epsilon_{ty_y} = 0.003 \times ((d - c_y) / c_y) = \mathbf{0.0307}$$

Pass - Ductility is sufficient at ultimate strength

Necessary shear checks

Beam action at distance 'd' in the section parallel to X axis is not applicable

Beam action at column face in the section parallel to X axis must be checked

Beam action at distance 'd' in the section parallel to Y axis is not applicable

Beam action at column face in the section parallel to Y axis must be checked

Punching of single pile is not applicable

Punching of pile at corner must be checked

Punching of overlapping piles must be checked

Punching shear at d / 2 from column face is not applicable

Punching shear at column face must be checked

Beam action at face of column

Dist. from face of column to edge of cap (X axis)

$$x_o = \max ((L_x / 2 - b_{col} / 2), 0 \text{ in}) = \mathbf{30.0 \text{ in}}$$

Dist. from face of column to edge of cap (Y axis)

$$y_o = \max ((L_y / 2 - h_{col} / 2), 0 \text{ in}) = \mathbf{30.0 \text{ in}}$$

Dist. from face of column to pile (X axis)

$$w_1 = \max (s / 2 - b_{col} / 2, 0.001 \text{ in}) = \mathbf{10.0 \text{ in}}$$

Dist. from face of column to pile (Y axis)

$$w_2 = \max (s / 2 - h_{col} / 2, 0.001 \text{ in}) = \mathbf{10.0 \text{ in}}$$

Shear at face of column parallel to X axis

$$V_{act_facex} = 2 \times P_{u_pile} - 1.2 \times (P_{wt} + P_{sur}) \times L_x \times ((L_y - h_{col}) / 2) / A_{cap}$$

$$V_{act_facex} = \mathbf{158.8 \text{ kips}}$$

Shear at face of column parallel to Y axis

$$V_{act_facey} = 2 \times P_{u_pile} - 1.2 \times (P_{wt} + P_{sur}) \times L_y \times ((L_x - h_{col}) / 2) / A_{cap}$$

$$V_{act_facey} = \mathbf{158.8 \text{ kips}}$$

$$VM_{rat1} = M_{ux} / (V_{act_facex} \times d) = \mathbf{0.3}$$

$$VM_{rat2} = M_{uy} / (V_{act_facey} \times d) = \mathbf{0.3}$$

$$\rho_{wx} = A_{st_x} / (L_y \times d) = \mathbf{0.004675}$$

$$f_1 = (d / w_1) \times (3.5 - 2.5 \times VM_{rat1}) = \mathbf{8.4}$$

$$f_2 = 1.9 \times 1 \text{ psi} \times \lambda \times \sqrt{(f'_c / 1 \text{ psi})} = \mathbf{127.5 \text{ psi}}$$

$$f_3 = 2500 \times \rho_{wx} \times 1 \text{ psi} / VM_{rat1} = \mathbf{37.8 \text{ psi}}$$

$$V_{c1} = f_1 \times (f_2 + f_3) \times L_x \times d = \mathbf{3275.9 \text{ kips}}$$

$$V_{c2} = \min (V_{c1}, 10 \times 1 \text{ psi} \times \lambda \times \sqrt{(f'_c / 1 \text{ psi})} \times L_x \times d) = \mathbf{1577.2 \text{ kips}}$$

$$V_c = \max (V_{c2}, 2 \times 1 \text{ psi} \times \lambda \times \sqrt{(f'_c / 1 \text{ psi})} \times L_x \times d) = \mathbf{1577.2 \text{ kips}}$$

$$V_s = 0 \text{ kips}$$

Shear resist. at face of column parallel to X axis

$$V_{r_facex} = \phi_v \times (V_c + V_s) = \mathbf{1182.9 \text{ kips}}$$

PASS - Pile cap is safe in beam action at face of column parallel to X axis

$$\rho_{wy} = A_{st_y} / (L_x \times d) = \mathbf{0.004675}$$

$$f_1 = (d / w_2) \times (3.5 - 2.5 \times VM_{rat2}) = \mathbf{8.4}$$

$$f_2 = 1.9 \times 1 \text{ psi} \times \lambda \times \sqrt{(f'_c / 1 \text{ psi})} = \mathbf{127.5 \text{ psi}}$$

$$f_3 = 2500 \times \rho_{wy} \times 1 \text{ psi} / VM_{rat2} = \mathbf{37.8 \text{ psi}}$$

$$V_{c1} = f_1 \times (f_2 + f_3) \times L_y \times d = \mathbf{3275.9 \text{ kips}}$$

$$V_{c2} = \min (V_{c1}, 10 \times 1 \text{ psi} \times \lambda \times \sqrt{(f'_c / 1 \text{ psi})} \times L_y \times d) = \mathbf{1577.2 \text{ kips}}$$

$$V_c = \max (V_{c2}, 2 \times 1 \text{ psi} \times \lambda \times \sqrt{(f'_c / 1 \text{ psi})} \times L_y \times d) = \mathbf{1577.2 \text{ kips}}$$

$$V_s = 0 \text{ kips}$$



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Shear resist. at face of column parallel to Y axis $V_{r_facey} = \phi_v \times (V_c + V_s) = 1182.9$ kips

PASS - Pile cap is safe in beam action at face of column parallel to Y axis

Corner shear check

Shear force acting

$$V_{act_diag} = P_{u_pile} = 85.9 \text{ kips}$$

Shear force resistance

$$V_{r_diag} = \phi_v \times 2 \times 1 \text{ psi} \times \lambda \times \sqrt{(f'_c / 1 \text{ psi})} \times 2 \times [\sqrt{(2) \times (d_{pile}/2 + e) + d_{pile}/2 + d}] \times d$$

$$V_{r_diag} = 399.8 \text{ kips}$$

PASS - Pile cap is safe in shear at corner

Punching of single pile at corner check

Perimeter of area at 'd / 2' from pile at corner

$$b_o = \pi \times (d_{pile} + d) / 4 + 2 \times (e + d_{pile} / 2) = 72.15 \text{ in}$$

Shear resistance of concrete

$$V_c = 4 \times 1 \text{ psi} \times \lambda \times \sqrt{(f'_c / 1 \text{ psi})} \times b_o \times d = 598.926 \text{ kips}$$

Shear resistance of steel

$$V_s = 0 \text{ kips}$$

Ultimate shear resistance

$$V_{r_corner} = \phi_v \times (V_c + V_s) = 449.194 \text{ kips}$$

Shear acting

$$V_{act_corner} = P_{u_pile} = 85.9 \text{ kips}$$

PASS - Resistance to pile punching at corner is sufficient

Punching of overlapping shear perimeter of piles check

Perimeter of overlapped area

$$b_o = \pi \times (d_{pile} + d) + 2 \times s = 200.60 \text{ in}$$

Shear resistance of concrete

$$V_c = 4 \times 1 \text{ psi} \times \lambda \times \sqrt{(f'_c / 1 \text{ psi})} \times b_o \times d = 1665.216 \text{ kips}$$

Shear resistance of steel

$$V_s = 0 \text{ kips}$$

Ultimate shear resistance

$$V_{r_overlap} = \phi_v \times (V_c + V_s) = 1248.912 \text{ kips}$$

Shear acting

$$V_{act_overlap} = 2 \times P_{u_pile} = 171.9 \text{ kips}$$

PASS - Resistance to punching of overlapping shear perimeter of piles is sufficient

Punching shear check at face of column

Length of shear perimeter d / 2 from column face

$$b_o = 2 \times (b_{col} + h_{col}) + 4 \times d = 187.744 \text{ in}$$

Length of shear perimeter at column face

$$b_s = 2 \times (b_{col} + h_{col}) = 64.000 \text{ in}$$

$$w = 10.000 \text{ in}$$

$$V_{act_pun_face} = P_u + (P_{sur} + P_{wt}) \times (s \times \min(L_y, s)) / L_x^2 = 316.811 \text{ kips}$$

$$V_{c1} = (d / w) \times (b_o / b_s) \times 2 \times 1 \text{ psi} \times \lambda \times \sqrt{(f'_c / 1 \text{ psi})} \times b_s \times d = 2410.630 \text{ kips}$$

$$V_{c2} = \min(V_{c1}, 32 \times 1 \text{ psi} \times \lambda \times \sqrt{(f'_c / 1 \text{ psi})} \times b_s \times d) = 2410.630 \text{ kips}$$

$$V_c = \max(V_{c2}, 4 \times 1 \text{ psi} \times \lambda \times \sqrt{(f'_c / 1 \text{ psi})} \times b_o \times d) = 2410.630 \text{ kips}$$

$$V_s = 0 \text{ kips}$$

$$V_{r_pun_face} = \phi_v \times (V_c + V_s) = 1807.973 \text{ kips}$$

PASS - Pile cap is safe in punching shear at face of column

Development length of tension steel (ACI 318, cl. 12.5)

All bars must be provided with standard 180° end hooks.

$$l_{dh_1} = (0.02 \times \psi_e \times f_y \times 1 \text{ psi}^{-1} / (\lambda_{dev} \times \sqrt{(f'_c / 1 \text{ psi})})) \times D_{x_bar} = 24.214 \text{ in}$$

Modification factor

$$\text{mod}_1 = 0.7$$

Required development length of hook

$$l_{dh_req} = \max(l_{dh_1} \times \text{mod}_1, 6 \text{ in}, 8 \times D_{x_bar}) = 16.950 \text{ in}$$

Provided development length

$$l_{dh_prov} = e - C_c + d_{pile} / 2 = 15.50 \text{ in}$$

FAIL - Hook development length is not sufficient